

Contour Integrals: Parametrization, Exactness, Winding, and Residues

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§1.1 A contour integral is a limit of transformed tangent increments

A contour is a piecewise C^1 path

$$\gamma : [a, b] \rightarrow \mathbb{C}.$$

For a continuous function f on the image of γ , define

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

The right-hand side is an ordinary complex-valued Riemann integral.

Let

$$a = t_0 < t_1 < \cdots < t_n = b$$

and choose $\tau_j \in [t_{j-1}, t_j]$. The Riemann sum is

$$\sum_{j=1}^n f(\gamma(\tau_j)) (\gamma(t_j) - \gamma(t_{j-1})).$$

Each directed increment is multiplied by f , hence scaled and rotated before the increments are added. A contour integral is not an area under a graph; it is a limit of transformed tangent vectors.

The basic estimate follows immediately:

$$\left| \int_{\gamma} f(z) dz \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| dt \leq L(\gamma) \max_{z \in \gamma} |f(z)|,$$

where

$$L(\gamma) = \int_a^b |\gamma'(t)| dt.$$

This ML -estimate is the main error-control tool for contour deformation and limiting arguments.

§1.2 Orientation, concatenation, and reparametrization

The reverse contour is

$$\bar{\gamma}(t) = \gamma(a + b - t).$$

Substitution gives

$$\boxed{\int_{\bar{\gamma}} f(z) dz = - \int_{\gamma} f(z) dz.}$$

If $\gamma_1(b_1) = \gamma_2(a_2)$, the concatenated contour $\gamma_1 * \gamma_2$ traverses the first and then the second. Splitting the parameter interval yields

$$\int_{\gamma_1 * \gamma_2} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz.$$

Let $\phi : [c, d] \rightarrow [a, b]$ be a piecewise C^1 , increasing surjection with $\phi(c) = a$, $\phi(d) = b$. Then

$$\begin{aligned} \int_{\gamma \circ \phi} f(z) dz &= \int_c^d f(\gamma(\phi(s))) \gamma'(\phi(s)) \phi'(s) ds \\ &= \int_a^b f(\gamma(t)) \gamma'(t) dt. \end{aligned}$$

Thus the integral is unchanged by orientation-preserving reparametrization. If ϕ reverses orientation, the sign changes.

For the unit circle, all three parametrizations

$$e^{it}, \quad 0 \leq t \leq 2\pi; \quad e^{2\pi it}, \quad 0 \leq t \leq 1; \quad e^{2\pi it^5}, \quad 0 \leq t \leq 1$$

give the same oriented contour integral.

§1.3 The complex integral is the pullback of a one-form

Write

$$z = x + iy, \quad f(z) = u(x, y) + iv(x, y), \quad dz = dx + i dy.$$

Then

$$\begin{aligned} f(z) dz &= (u + iv)(dx + i dy) \\ &= (u dx - v dy) + i(v dx + u dy). \end{aligned}$$

Therefore

$$\int_{\gamma} f(z) dz = \int_{\gamma} (u dx - v dy) + i \int_{\gamma} (v dx + u dy).$$

A complex contour integral is two real line integrals coupled by the complex multiplication rule.

In differential-form notation, $f(z) dz$ is a complex-valued one-form. The parametrized formula is exactly pullback:

$$\int_{\gamma} f(z) dz = \int_{[a,b]} \gamma^*(f(z) dz).$$

This explains reparametrization invariance: integration is applied to the pulled-back one-form on the parameter interval.

§1.4 Primitives are equivalent to path independence

Let $D \subseteq \mathbb{C}$ be a domain and let $f : D \rightarrow \mathbb{C}$ be continuous.

Theorem 1.4.1

The following are equivalent.

- (i) There is a holomorphic function $F : D \rightarrow \mathbb{C}$ with $F' = f$.
- (ii) The integral of f depends only on the endpoints of the contour.
- (iii) Every closed contour γ in D satisfies

$$\int_{\gamma} f(z) dz = 0.$$

Proof. If $F' = f$, the chain rule gives

$$\frac{d}{dt} F(\gamma(t)) = f(\gamma(t)) \gamma'(t).$$

Hence

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)),$$

which proves (i) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (iii) is immediate.

Assume (iii). Fix $z_0 \in D$, and for $z \in D$ define

$$F(z) = \int_{\gamma_{z_0, z}} f(\zeta) d\zeta,$$

where $\gamma_{z_0, z}$ is any contour from z_0 to z . If two contours are chosen, following one and reversing the other gives a closed contour, so (iii) makes F well defined.

For small h , use the same contour to z and then the line segment from z to $z + h$:

$$\frac{F(z+h) - F(z)}{h} = \int_0^1 f(z+th) dt.$$

By continuity, the right-hand side tends to $f(z)$. Thus $F'(z) = f(z)$. □

The topology of the domain enters in condition (iii): local primitives may exist while a global primitive fails because closed loops can carry nonzero periods.

§1.5 Cauchy–Goursat turns holomorphicity into exactness on simple domains

Suppose $f = u + iv$ has continuous first partial derivatives on a region containing a positively oriented simple closed contour $\gamma = \partial\Omega$. Using the decomposition of $f dz$ and Green's theorem,

$$\begin{aligned} \operatorname{Re} \int_{\gamma} f(z) dz &= \int_{\gamma} u dx - v dy \\ &= \iint_{\Omega} (-v_x - u_y) dx dy, \end{aligned}$$

and

$$\begin{aligned} \operatorname{Im} \int_{\gamma} f(z) dz &= \int_{\gamma} v dx + u dy \\ &= \iint_{\Omega} (u_x - v_y) dx dy. \end{aligned}$$

The Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x$$

make both integrands zero. Therefore

$$\int_{\gamma} f(z) dz = 0.$$

Goursat’s theorem removes the assumption of continuous partial derivatives: complex differentiability alone is enough. On a simply connected domain, every closed contour is deformable through the domain to a point, and Cauchy–Goursat implies

$$\int_{\gamma} f(z) dz = 0.$$

By the preceding theorem, every holomorphic function on a simply connected domain has a primitive.

The conclusion fails when the domain contains a hole around a singularity. The model obstruction is $1/z$ on $\mathbb{C}^*$.

§1.6 Powers on a circle isolate one Fourier mode

For the circle traversed k times,

$$\gamma_k(t) = e^{ikt}, \quad 0 \leq t \leq 2\pi,$$

one has

$$\begin{aligned} \int_{\gamma_k} z^m dz &= \int_0^{2\pi} e^{imkt} ik e^{ikt} dt \\ &= ik \int_0^{2\pi} e^{ik(m+1)t} dt. \end{aligned}$$

If $m \neq -1$, the oscillation completes an integer number of periods and cancels. If $m = -1$, the integrand is the constant ik . Hence

$$\boxed{\int_{\gamma_k} z^m dz = \begin{cases} 2\pi ik, & m = -1, \\ 0, & m \neq -1. \end{cases}}$$

The exceptional exponent -1 is the coefficient that survives contour averaging. This is the local algebra behind residues.

§1.7 Winding number is the period of $dz/(z - a)$

Let γ be a closed contour avoiding a . Define

$$\text{Ind}(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - a}.$$

For a circle centered at a traversed k times, the preceding computation gives

$$\text{Ind}(\gamma, a) = k.$$

In general the index is an integer, equal to the endpoint change of a lifted argument of $\gamma(t) - a$.

It is invariant under homotopies that avoid a . For a smooth homotopy $\Gamma(t, s)$, differentiate under the integral sign:

$$\begin{aligned} \frac{d}{ds} \int_0^1 \frac{\partial_t \Gamma(t, s)}{\Gamma(t, s) - a} dt &= \int_0^1 \partial_t \left(\frac{\partial_s \Gamma(t, s)}{\Gamma(t, s) - a} \right) dt \\ &= 0, \end{aligned}$$

because the loop has the same initial and terminal point. Approximation or covering-space lifting gives the general continuous-homotopy statement.

Thus the integral depends on the homotopy class of the loop in $\mathbb{C} \setminus \{a\}$, not on its detailed shape.

§1.8 Closed need not mean exact on a punctured domain

On $\mathbb{R}^2 \setminus \{0\}$, consider the real one-form

$$\alpha = \frac{1}{2\pi} \frac{-y dx + x dy}{x^2 + y^2}.$$

A direct calculation gives

$$d\alpha = 0.$$

So α is closed. On the unit circle $x = \cos t$, $y = \sin t$,

$$\alpha = \frac{dt}{2\pi},$$

and therefore

$$\int_{S^1} \alpha = 1.$$

If $\alpha = dF$ were exact, every closed-loop integral would vanish. Hence α is closed but not exact.

This period represents the generator of

$$H_{\text{dR}}^1(\mathbb{C}^*) \cong \mathbb{R}.$$

The complex form dz/z contains the same obstruction:

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z} = \text{Ind}(\gamma, 0).$$

The failure of a global logarithm on $\mathbb{C}^*$ is the same phenomenon. Locally,

$$d(\log z) = \frac{dz}{z},$$

but no single-valued logarithm exists around a loop of nonzero winding.

§1.9 Cauchy's integral formula recovers interior values from the boundary

Let f be holomorphic on a neighborhood of a positively oriented simple closed contour γ and its interior, and let z lie inside. Then

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

To see the mechanism, write

$$\frac{f(\zeta)}{\zeta - z} = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}.$$

The first term has a removable singularity at $\zeta = z$, with value $f'(z)$, so its closed-contour integral vanishes. The second term contributes

$$f(z) \int_{\gamma} \frac{d\zeta}{\zeta - z} = 2\pi i f(z) \text{Ind}(\gamma, z).$$

For a simple positively oriented contour containing z , the index is 1.

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

For example,

$$\int_{|\zeta|=2} \frac{e^{\zeta}}{(\zeta - 1)^3} d\zeta = \frac{2\pi i}{2!} e = \pi i e.$$

§1.10 Residues convert local Laurent coefficients into global integrals

If f has an isolated singularity at a , its Laurent expansion is

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n.$$

The residue is

$$\text{Res}(f, a) = c_{-1}.$$

All other powers integrate to zero around a small circle, while $(z - a)^{-1}$ contributes $2\pi i$. Therefore

$$\int_{|z-a|=r} f(z) dz = 2\pi i \text{Res}(f, a).$$

For a contour γ avoiding finitely many poles $a_1, \dots, a_N$, the residue theorem is

$$\boxed{\int_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^N \text{Ind}(\gamma, a_j) \text{Res}(f, a_j).}$$

Local Laurent coefficients are paired with global winding numbers.

Consider

$$f(z) = \frac{z + 1}{z(z - 1)}.$$

The residues are

$$\text{Res}(f, 0) = -1, \quad \text{Res}(f, 1) = 2.$$

Hence

$$\int_{|z|=2} f(z) dz = 2\pi i(-1 + 2) = 2\pi i.$$

For the smaller contour $|z| = 1/2$, only the pole at 0 is enclosed, so

$$\int_{|z|=1/2} f(z) dz = -2\pi i.$$

The integrand has not changed; the answer changes because the winding data relative to the poles has changed.